

4 Lung epithelial cells have a thin layer of watery mucus on their surface.

The normal allele of the *CFTR* gene codes for a transport protein that transports chloride ions out of epithelial cells.

Fig. 4.1 is a diagram of part of the cell surface membrane and the mucus layer of an epithelial cell with normal CFTR proteins.

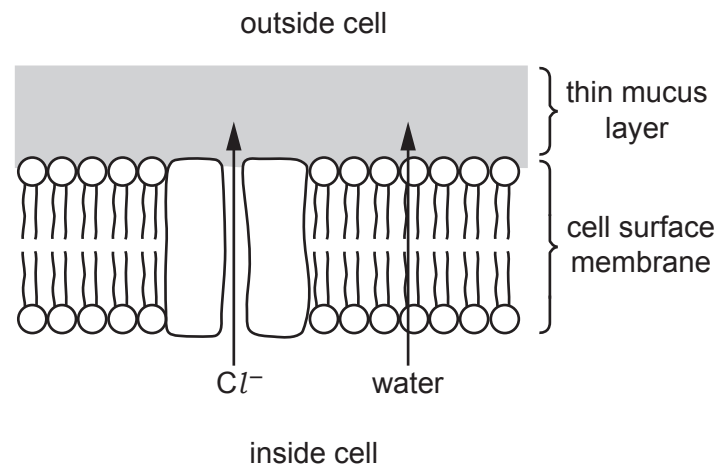


Fig. 4.1

Cystic fibrosis (CF) is a genetic disorder caused by having two recessive alleles of *CFTR*. In severe cases of CF, the transport proteins are not added to the cell surface membrane. This causes the mucus layer to be thick and sticky.

(a) Explain why the absence of CFTR proteins will cause the mucus layer to be thick and sticky.

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- (b) The probability of a baby having CF when both parents are heterozygous carriers for CF is 25%.

It is possible to carry out prenatal screening to check for CF by using one of these tests:

- amniocentesis, using cells from the amniotic fluid
- chorionic villus sampling, using cells from the placenta.

Both tests slightly increase the probability of the pregnancy failing (miscarriage).

Outline the advantages of carrying out prenatal screening for CF.

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- (c) Embryos produced by IVF may be screened for genetic abnormalities:
- to test for a specific genetic disease, such as cystic fibrosis
 - to check whether there is an abnormal number of chromosomes present.

To improve the success of implantation and pregnancy, only embryos without any form of genetic abnormality are transferred to the woman's uterus.

A new double screening method was trialled where a single embryo biopsy was taken and used to test for a specific genetic disease **and** to check the number of chromosomes. In the trial, 1122 embryos were tested using this double screening method.

In the trial, of the 1122 embryos tested:

- 50.6% did not have a genetic disease
- 27.5% did not have a genetic disease **and** did not have an abnormal number of chromosomes (normal embryos).

Only normal embryos were transferred into the women. The percentage of embryo transfers that resulted in pregnancy was calculated.

The results of the trial using double screening of a single biopsy were compared to the results of IVF procedures that used standard screening methods, as shown in Table 4.1.

Table 4.1

IVF method	percentage of embryo transfers that resulted in pregnancy
IVF with standard screening	32
IVF with double screening	49

Using the data in Table 4.1, discuss the social **and** ethical considerations of double screening for cystic fibrosis and chromosomal abnormalities in a single biopsy.

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